

Problem Statement — Swachhata Abhiyan Digital Platform

Background

Swachh Bharat Abhiyan (Clean India Mission) is a national movement aimed at improving sanitation, reducing waste, and promoting cleanliness in urban and rural areas. Currently, citizen feedback, waste collection tracking, and cleanliness monitoring are handled by **multiple disconnected systems, manual reporting, and delayed communication.**

This results in:

- **Delayed grievance resolution** for waste management issues.
- **Lack of transparency** in cleanliness drives and fund usage.
- **Difficulty in tracking field staff performance** and resource allocation.
- **Limited citizen participation** due to no central digital platform.
- **Poor analytics** to measure progress and identify problem zones.

Need

To address these gaps, a **Cloud-based Swachhata Abhiyan Management Platform** will be developed that will:

- **Centralize citizen complaints, cleanliness drives, and waste management data.**
- **Enable real-time issue reporting via mobile app** (geo-tagged images, location-based complaints).
- **Track waste collection vehicles & workforce using IoT & GPS.**
- **Provide dashboards** for municipal officers to monitor progress.
- **Encourage citizen engagement** through gamification, rewards, and feedback loops.
- **Leverage AI analytics** to identify high-waste areas and optimize cleaning schedules.

Objectives

1. **Citizen Engagement** – Mobile/web platform for reporting cleanliness issues with geo-tagged images.
2. **Workforce & Vehicle Tracking** – GPS-enabled real-time monitoring of waste collection teams.
3. **Resource Allocation** – AI-based route optimization for waste collection trucks.

4. **Progress Monitoring** – Public dashboards to showcase cleanliness scores per ward/city.
 5. **Feedback & Gamification** – Reward citizens and wards for maintaining cleanliness.
 6. **Scalable Infrastructure** – Cloud-hosted platform with nationwide reach.
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Agile Delivery Plan

Epics & Features

1. **Citizen Complaint System**
 - Complaint submission with image & GPS location
 - Issue categorization (garbage, street cleaning, toilet maintenance, etc.)
 - Auto-assign to nearest field team
 2. **Waste Collection Management**
 - GPS-based vehicle tracking
 - Real-time route monitoring & optimization
 - Work completion verification via mobile app
 3. **Cleanliness Drive Management**
 - Event scheduling & volunteer registration
 - Progress tracking and live updates
 - Digital certificates for participants
 4. **Data & Analytics**
 - City & ward cleanliness scorecards
 - AI-based hotspot detection
 - Predictive analytics for resource planning
 5. **Citizen Engagement & Rewards**
 - Leaderboards for cleanliness ratings
 - Reward points for complaint resolution & participation
 - Public awareness campaigns through app
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Sample Sprint Plan (2-week sprints)

Sprint Deliverables

- 1 Cloud infrastructure setup, basic complaint submission API
- 2 Geo-tagged image uploads, complaint tracking UI
- 3 GPS vehicle tracking, route mapping
- 4 Volunteer registration & cleanliness event module
- 5 Analytics dashboard with cleanliness scores
- 6 AI-based waste hotspot detection
- 7 Reward & gamification system
- 8 Final integration, UAT, public launch

Latest DevOps Toolchain

Stage	Tools
Project Management	Jira, Confluence
Source Control	GitHub
CI/CD	GitHub Actions, ArgoCD
Testing	Cypress (UI), Postman (API), PyTest
Containerization	Docker
Orchestration	Kubernetes (EKS/GKE)
Monitoring	Prometheus, Grafana, AWS CloudWatch
Logging	ELK Stack (Elastic, Logstash, Kibana)
Security	SonarQube, OWASP ZAP, Snyk
Infrastructure as Code	Terraform

Cloud Architecture (AWS Example)

- **Frontend:** React.js / Flutter → AWS Amplify
 - **Backend:** Node.js / Django → AWS Lambda / EKS
 - **Database:** Amazon RDS (PostgreSQL) + Amazon DynamoDB (fast lookups)
 - **Storage:** Amazon S3 (for images, documents)
 - **AI/ML:** AWS SageMaker (hotspot prediction)
 - **Auth:** AWS Cognito
 - **Notifications:** AWS SNS + Twilio
 - **CI/CD:** AWS CodePipeline + GitHub Actions
 - **Monitoring:** Prometheus + AWS CloudWatch
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Workflow

1. **Backlog Planning** → Jira Epics & User Stories.
 2. **Development** → Code in GitHub (feature branches).
 3. **Build & Test** → Automated in GitHub Actions + PyTest + Cypress.
 4. **Containerization** → Docker Images pushed to AWS ECR.
 5. **Deployment** → ArgoCD for GitOps deployment to Kubernetes.
 6. **Monitoring & Feedback** → Real-time dashboards & sprint reviews.
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